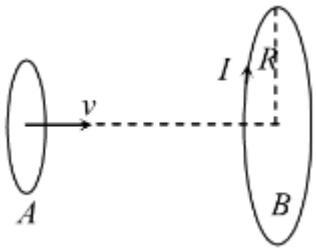
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Loop A of radius $r(r < R)$ moves towards a constant current carrying loop B with a constant velocity v in such a way that their planes are parallel and coaxial. The distance between the loops when the induced emf in loop A is maximum is



(A) R

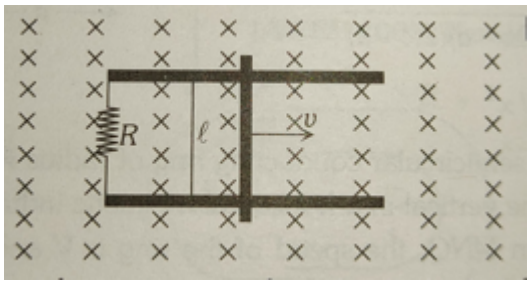
(B) $\frac{R}{\sqrt{2}}$

(C) $\frac{R}{2}$

(D) $R\left(1-\frac{1}{\sqrt{2}}\right)$

Q.2

A conducting bar of mass m and length ℓ moves on two frictionless parallel rails in the presence of a constant uniform magnetic field of magnitude B directed into the page as shown in the figure. The bar is given an initial velocity v_0 towards the right at $t = 0$. Then the



(A) Induced current in the circuit is in the clockwise direction

(B) Velocity of the bar decreases linearly with time

(C) Distance the bar travels before it comes to a complete stop is proportional to R

(D) Power generated across the resistance is proportional to l

Q.3

5.5×10^{-4} magnetic flux lines are passing through a coil of resistance 10 ohm and number of turns 1000 . If the number of flux lines reduces to 5×10^{-5} in 0.1sec. The electromotive force and the current induced in the coil will be respectively-

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(A) $5\ \text{V}, 0.5\ \text{A}$

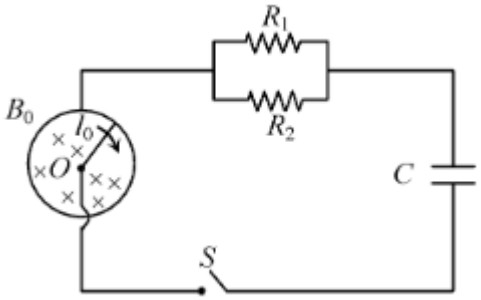
(B) $5 \times 10^{-4}\ \text{V}, 5 \times 10^{-4}\ \text{A}$

(C) $50\ \text{V}, 5\ \text{A}$

(D) none of the above

Q.4

There is a small metallic ring of radius l_0 and having negligible resistance placed perpendicular to a constant magnetic field B_0. One end of a rod is hinged at the centre of ring O and other end is placed on the ring. Now rod is rotated with constant angular velocity ω_0 by some external agent and circuit is connected as shown in the figure, initially switch is open and capacitor is uncharged. If switch S is closed at $t = 0$, then calculate heat loss from the resistor R_2 from $t = 0$ to the instant when voltage across the capacitor becomes V_0. (Assume plane of ring to be horizontal and friction to be an absent at all the contacts).
(Assume, $R_2 = 2R_1, \quad B_0 l_0^2 \omega_0 = 4V_0$)



(A) $\frac{1}{2}CV_0^2$

(B) $\frac{1}{6}CV_0^2$

(C) $\frac{2}{3}CV_0^2$

(D) $\frac{1}{3}CV_0^2$

Q.5

A closed coil consists of 500 turns on a rectangular frame of area $4.0\ \text{cm}^2$ and has a resistance of 50 ohm. The coil is kept with its plane perpendicular to a uniform magnetic field of 0.2 weber/meter ^{2}. The amount of charge flowing through the coil if it is turned over (rotated through 180°) will be -

(A) $1.6 \times 10^{-19}\text{C}$

(B) $1.6 \times 10^{-9}\text{C}$

(C) $1.6 \times 10^{-3}\text{C}$

(D) $1.6 \times 10^{-2}\text{C}$

Q.6

A very small circular loop of area $5 \times 10^{-4}\ \text{m}^2$, resistance 2ohm and negligible inductance is initially coplanar and concentric with a much larger fixed circular loop of radius $0.1\ \text{m}$. A constant current of 1 ampere is passed in a bigger loop and the smaller loop is rotated with angular velocity $\omega\text{rad/sec}$ about a diameter. Calculate the flux

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linked with the smaller loop.
$\left(\mu_0 = 4\pi \times 10^{-7}\ \mathrm{V-s/A-m}\right)$.

(A) $\pi \times 10^{-9} \cos \omega t$ weber.

(B) $\pi \times 10^{-6} \cos \omega t$ weber.

(C) $2\pi \times 10^{-4} \cos \omega \mathrm{t}$ weber.

(D) $\pi \times 10^{-9} \sin \omega t$ weber.

Q.7

A coil of wire having finite inductance and resistance has a conducting ring placed coaxially within it. The coil is connected to a battery at time $t = 0$, so that a time-dependent current $I_1(t)$ starts flowing through the coil. If $I_2(t)$ is the current induced in the ring and $B(t)$ is the magnetic field at the axis of the coil due to $I_1(t)$, then as a function of time $(t > 0)$, the product $I_2(t)B(t)$

(A) Increases with time

(B) Decreases with time

(C) Does not vary with time

(D) Passes through a maximum

Q.8

Two rail tracks, insulated from each other and the ground, are connected to millivoltmeter. What is the reading of the milli voltmeter when a train passes at a speed of $180\ \mathrm{km/hr}$ along the track, given that the horizontal component of earth's magnetic field is $0.2 \times 10^{-4}\ \mathrm{Wb/m^2}$ and rails are separated by 1 meter.

(A) $1\mathrm{mV}$

(B) $10\mathrm{mV}$

(C) $100\mathrm{mV}$

(D) $1\ \mathrm{V}$

Q.9

The magnetic flux ϕ linked with a coil depends on time as $\phi = \mathrm{Kt^m}$ where K and m are constants. If the emf induced in the coil is ε, then -

(A) $|\varepsilon|$ increases with t for $m > 1$

(B) $|\varepsilon|$ decreases with t for $0 < \mathrm{m} < 1$

(C) for $\mathrm{m} = 1$, $|\varepsilon|$ is infinite

(D) for $0 < \mathrm{m} < 1$, $|\varepsilon|$ is zero

Q.10

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A circular coil expands radially in a region of magnetic field and no electromotive force is produced in the coil. This can be because

(A) The magnetic field is constant

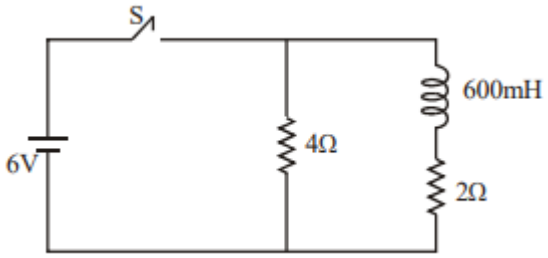
(B) The magnetic field is in the same plane as the circular coil and it may or may not vary

(C) The magnetic field has a perpendicular (to the plane of the coil) component whose magnitude is decreasing suitably

(D) There is a constant magnetic field in the perpendicular (to the plane of the coil) direction

Q.11

A 600mH inductor and two resistors 4Ω and 2Ω are connected to a cell of emf 6V as shown in the figure. Select true statements from the following -



(A) Just after closing the switch S, no current flows in 2Ω.

(B) After a long time of closing the switch S, the current in 4Ω is $1.5\ \text{A}$.

(C) When the switch S is opened, the time constant of the circuit is $0.45\ \text{s}$.

(D) Just after opening the switch S, the current in 2Ω is $3.0\ \text{A}$.

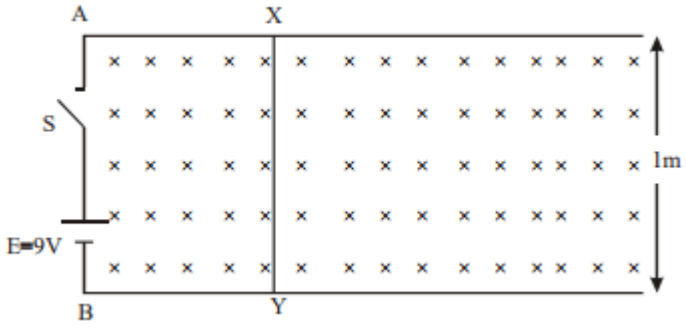
Q.12

Passage

A conducting rod of length $\ell = 1\ \text{m}$ can move without friction on two very long (say, infinitely long) conducting horizontal rails as shown. Mass of the rod is $m = 1\ \text{kg}$ and its resistance $R = 3\Omega$. Let the system of the rails and the rod be in the plane of paper. A uniform magnetic field $B = 0.8\ T$ is applied perpendicular to the plane of paper and directed into it. At one end, the rails are connected through a battery of emf $E = 9\ \text{V}$ and a switch S. Except for the resistance of rod, all other resistances in the circuit can be neglected. The switch is closed at $t = 0$. Consequently, current passes through the rod and it experiences a force due to magnetic field and moves to the right. Motion of the rod results in change of area AXYB and hence a change of magnetic flux. Induced emf is developed and induced current appears in the section AXYB. Direction of induced current can be determined using Lenz's law and it can be made out that induced current will oppose the current due to the external source of emf. In other words, force acting on the rod towards right decreases. So the acceleration decreases and finally becomes zero. Velocity of the rod when acceleration becomes zero so that the rod then continues to move at same velocity is called Terminal

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Velocity.



Question

Just after the switch is closed at $t = 0$, acceleration of the rod is -

(A) $4.6\ \mathrm{m/s^2}$

(B) $3.8\ \mathrm{m/s^2}$

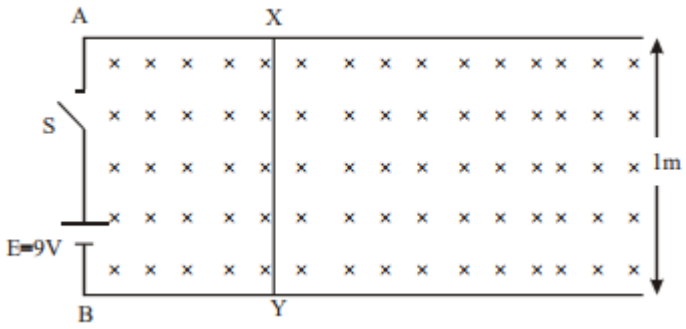
(C) $3.2\ \mathrm{m/s^2}$

(D) $2.4\ \mathrm{m/s^2}$

Q.13

Passage

A conducting rod of length $\ell = 1\ \mathrm{m}$ can move without friction on two very long (say, infinitely long) conducting horizontal rails as shown. Mass of the rod is $m = 1\ \mathrm{kg}$ and its resistance $R = 3\Omega.$ Let the system of the rails and the rod be in the plane of paper. A uniform magnetic field $B = 0.8\ T$ is applied perpendicular to the plane of paper and directed into it. At one end, the rails are connected through a battery of emf $E = 9\ \mathrm{V}$ and a switch S. Except for the resistance of rod, all other resistances in the circuit can be neglected. The switch is closed at $t = 0.$ Consequently, current passes through the rod and it experiences a force due to magnetic field and moves to the right. Motion of the rod results in change of area AXYB and hence a change of magnetic flux. Induced emf is developed and induced current appears in the section AXYB. Direction of induced current can be determined using Lenz's law and it can be made out that induced current will oppose the current due to the external source of emf. In other words, force acting on the rod towards right decreases. So the acceleration decreases and finally becomes zero. Velocity of the rod when acceleration becomes zero so that the rod then continues to move at same velocity is called Terminal Velocity.



Question

Acceleration of the rod when its speed is $6\mathrm{m/s}$ will be nearly

(A) $1.1\ \mathrm{m/s^2}$

(B) $2.3\ \mathrm{m/s^2}$

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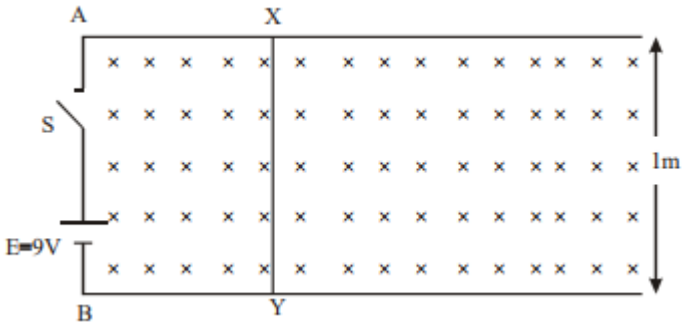
(C) $3.7\ \mathrm{m/s^2}$

(D) $4.3\ \mathrm{m/s^2}$

Q.14

Passage

A conducting rod of length $\ell = 1\ \mathrm{m}$ can move without friction on two very long (say, infinitely long) conducting horizontal rails as shown. Mass of the rod is $m = 1\ \mathrm{kg}$ and its resistance $R = 3\Omega.$ Let the system of the rails and the rod be in the plane of paper. A uniform magnetic field $B = 0.8\ T$ is applied perpendicular to the plane of paper and directed into it. At one end, the rails are connected through a battery of emf $E = 9\ \mathrm{V}$ and a switch S. Except for the resistance of rod, all other resistances in the circuit can be neglected. The switch is closed at $t = 0.$ Consequently, current passes through the rod and it experiences a force due to magnetic field and moves to the right. Motion of the rod results in change of area AXYB and hence a change of magnetic flux. Induced emf is developed and induced current appears in the section AXYB. Direction of induced current can be determined using Lenz's law and it can be made out that induced current will oppose the current due to the external source of emf. In other words, force acting on the rod towards right decreases. So the acceleration decreases and finally becomes zero. Velocity of the rod when acceleration becomes zero so that the rod then continues to move at same velocity is called Terminal Velocity.



Question

Terminal speed of the rod is -

(A) $18.75\ \mathrm{m/s}$

(B) $13.5\ \mathrm{m/s}$

(C) $11.25\ \mathrm{m/s}$

(D) $8.75\ \mathrm{m/s}$

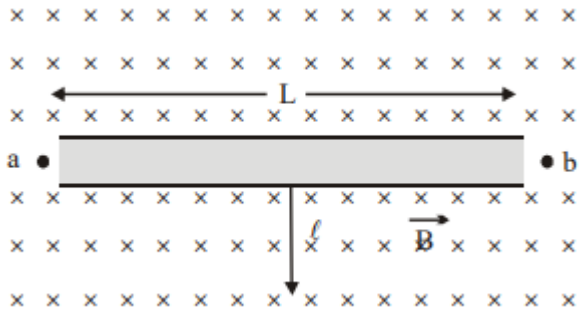
Q.15

Passage

In figure, a conducting rod of length $\mathrm{L} = 30\ \mathrm{cm}$ moves in a magnetic field of magnitude $0.450\ \mathrm{T}$ directed into

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the plane of the figure. The rod moves with speed $v = 5.00\ m/s$ in the direction shown.



Question

What is motional emf induced in the rod ?

(A) $0.675\ V$

(B) $0.175\ V$

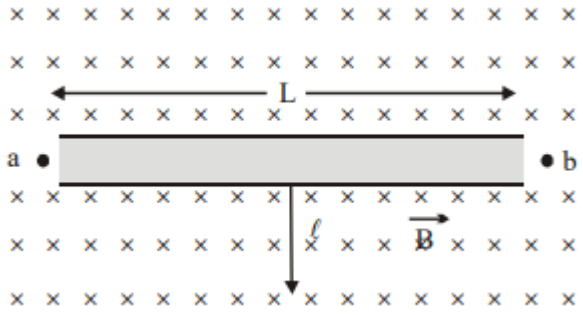
(C) $0.275\ V$

(D) $0.975\ V$

Q.16

Passage

In figure, a conducting rod of length $L = 30\ cm$ moves in a magnetic field of magnitude $0.450\ T$ directed into the plane of the figure. The rod moves with speed $v = 5.00\ m/s$ in the direction shown.



Question

When the charges in the rod are in equilibrium, what is the magnitude of the electric field within the rod ?

(A) $1.25\ V/m$

(B) $3.25\ V/m$

(C) $2.25\ V/m$

(D) $4.25\ V/m$

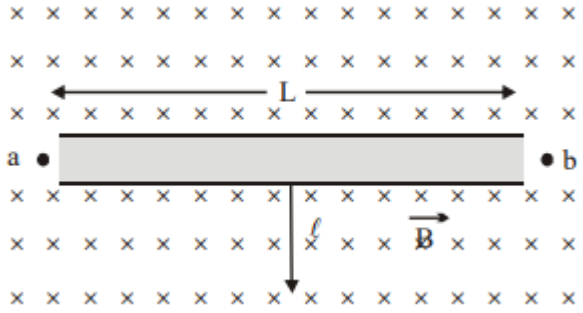
Q.17

Passage

In figure, a conducting rod of length $L = 30\ cm$ moves in a magnetic field of magnitude $0.450\ T$ directed into

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the plane of the figure. The rod moves with speed $v = 5.00\ m/s$ in the direction shown.



Question

Choose the correct statement -

(A) b is at higher potential

(B) b has an excess of positive charge

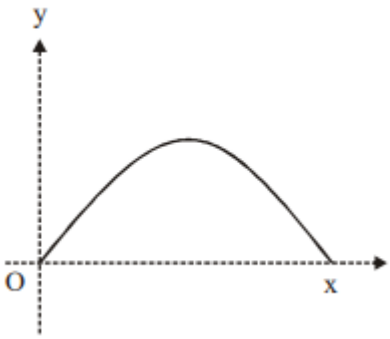
(C) a is at higher potential

(D) Both (A) and (B)

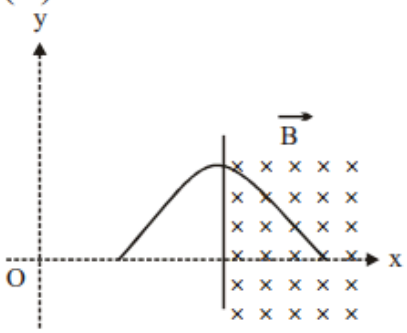
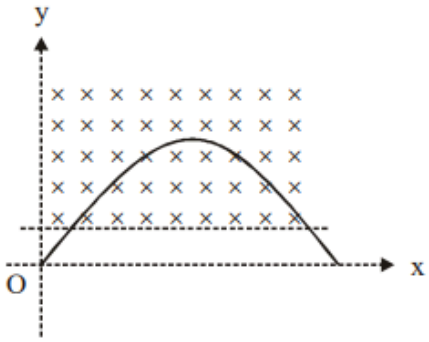
Q.18

A conducting loop is on xy plane as shown at time $t = 0$, in a region of uniform magnetic field directed towards z− axis $y = a \sin bx$

The motion of the loop and region of magnetic field is as given in column I. Match with column II.



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Column I	Column II (Till steady state is reached)
(A)  $\vec{B}$ is in restricted region as shown. Loop is moving with constant velocity towards +x direction.	(p) plot of induced emf vs time is sinusoidal.
(B)  $\vec{B}$ is in restricted region as shown. Loop is moving with constant velocity towards +x direction	(q) plot of induced emf vs time is not sinusoidal.
(C) $\vec{B}$ is in entire region and loop is rotating about x-axis at a constant angular velocity b	(r) The maximum value of induced emf is independent of a.
(D) $\vec{B}$ is in entire region but varies with time as per $B = B_0 \sin bt$, and loop is stationary.	(s) The maximum value of individual emf is independent of b.

(A) $(A) - s(B) - p, r, s(C) - q, s(D) - p, r$

(B) $(A) - q, s(B) - p, q(C) - p, q(D) - r, s$

(C) $(A) - p, r(B) - q(C) - q, s(D) - p$

(D) $(A) - p, s, (B) - q, r, s(C) - p, s(D) - p, s$

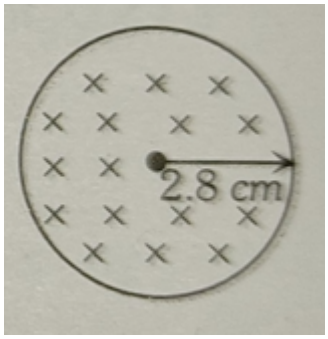
Q.19

A long coaxial cable consists of two thin-walled conducting cylinders with inner radius $2\ \text{cm}$ and outer radius $8\ \text{cm}$. The inner cylinder carries a steady current $0.1\ \text{A}$, and the outer cylinder provides the return path for that current. The current produces a magnetic field between the two cylinders. Find the energy stored in the magnetic field for length $5\ \text{m}$ of the cable. Express answer in nJ (use $\ln 2 = 0.7$)

Q.20

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The magnetic field of a cylindrical magnet that has a pole face radius $2.8\ \text{cm}$ can be varied sinusoidally between the minimum value $16.8\ \text{T}$ and the maximum value $17.2\ \text{T}$ at a frequency of $50/\pi\text{Hz}$. Cross section of the magnetic field created by the magnet is shown in figure. At a radial distance of $2\ \text{cm}$ from the axis, find the amplitude of the electric field (in units of $\times 10^2 \text{mNC}^{-1}$) induced by the magnetic field variation



Answers & Solutions

Q.1 Answer:

$\frac{R}{2}$

Solution:

Magnetic flux through the loop A;

$\phi = B\pi r^2 = \frac{\mu_0 I R^2}{2(R^2+x^2)^{3/2}}\pi r^2$

Induced emf in the loop A;

$\varepsilon = -\frac{d\phi}{dt} = \frac{\mu_0 I R^2 \pi r^2}{2}\left(-\frac{3}{2(R^2+x^2)^{5/2}}2x\frac{dx}{dt}\right)$

$= -\frac{3\mu_0 I R^2 \pi r^2 v}{2}\left[\frac{x}{(R^2+x^2)^{5/2}}\right]$

Induced emf is maximum when $\frac{d\varepsilon}{dx} = 0$

$\left(\mathrm{R}^2 + \mathrm{x}^2\right) - 5\mathrm{x}^2 = 0$

$\mathrm{R}^2 = 4\mathrm{x}^2$

$\mathrm{x}^2 = \frac{\left(\mathrm{R}^2\right)}{4}$

$\therefore \mathrm{x} = \frac{\mathrm{R}}{2}$

$\therefore$ option (3) is correct

Q.2 Answer:

Distance the bar travels before it comes to a complete stop is proportional to R

Solution:

$F = iB\ell \quad \Rightarrow \vec{a} = \frac{iB\ell}{m}$

$\phi = BA \quad \Rightarrow \frac{d\phi}{dt} = Bl\frac{dx}{dt}$

$e = Blv$

$i = \frac{e}{R} = \frac{Blv}{R}$

By eq. (i) and (ii)

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$$a = \left(\frac{Blv}{R}\right)\frac{Bl}{m} = \frac{B^2l^2v}{Rm}$$

$$a = v \cdot \frac{dv}{dx} = \frac{B^2l^2v}{Rm}$$

$$dv = \frac{B^2l^2}{Rm}\int dx \Rightarrow v = \frac{B^2l^2x}{Rm} \Rightarrow x = \frac{Rmv}{B^2l^2}.$$

Q.3 Answer:

$5\ \text{V}, 0.5\ \text{A}$

Solution:

Initial magnetic flux $\phi_1 = 5.5 \times 10^{-4}$ **weber.**

Final magnetic flux $\phi_2 = 5 \times 10^{-5}$ **weber.**

$\therefore$ **Change in flux** $\Delta\phi = \phi_2 - \phi_1 = \left(5 \times 10^{-5}\right) - \left(5.5 \times 10^{-4}\right) = -50 \times 10^{-5}$ **weber.**

Time interval for this change, $\Delta \mathrm{t} = 0.1\mathrm{sec}.$

$\therefore$ **Induced emf in the coil** $\mathrm{e} = -\mathrm{N}\frac{\Delta\phi}{\Delta \mathrm{t}} = -1000 \times \frac{\left(-50\times10^{-5}\right)}{0.1} = 5\mathrm{volt}.$

Resistance of the coil, $\mathrm{R} = 10\mathrm{ohm}.$

Hence induced current in the coil is $\mathrm{i} = \frac{\mathrm{e}}{\mathrm{R}} = \frac{5\mathrm{volt}}{10\mathrm{ohm}} = 0.5$ **ampere.**

Q.4 Answer:

$\frac{1}{2}CV_0^2$

Solution:

$$\text{Voltage across rod } = \frac{1}{2}B_0l_0^2\omega_0$$

$$\text{Charge on capacitor } = CV_0$$

$$v \times q = \frac{1}{2}CV_0^2 + H_{R_1} + H_{R_2}$$

$$CV_0 \times \frac{1}{2}B_0l_0^2\omega_0 = \frac{1}{2}CV_0^2 + \frac{R_2}{R_1}H_{R_2} + H_{R_2}$$

$$\frac{1}{2}CV_0B_0l_0^2\omega_0 - \frac{1}{2}CV_0^2 = \frac{H_{R_2}\left[R_2 + R_1\right]}{R_1} =$$

$$H_{R_2} = \frac{R_1}{R_1 + R_2}\left[\frac{1}{2}CV_0B_0l_0^2\omega_0 - \frac{1}{2}CV_0^2\right] = \frac{R_1}{R_1 + R_2} \times \frac{1}{2}CV_0\left[B_0l_0^2\omega_0 - V_0\right]$$

$$= \frac{1}{2}CV_0^2$$

Q.5 Answer:

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$1.6 \times 10^{-3}\mathrm{C}$

Solution:

The magnetic flux passing through each turn of a coil of area A, perpendicular to a magnetic field B is given by $\phi_1 = \mathrm{BA}$.

The magnetic flux through it on rotating it through 180° will be $\phi_2 = -\mathrm{BA}$.($-$ sign is put because now the flux lines enters the coils through the outer face)

$\therefore$ **Change in magnetic flux $\Delta\phi = \phi_1 - \phi_2 = -\mathrm{BA} - (\mathrm{BA}) = -2\mathrm{BA}$.**

Suppose this change takes in time Δt. According to Faraday's law, the emf induced in the coil is given by $\mathrm{e} = -\mathrm{N}\frac{\Delta\phi}{\Delta \mathrm{t}} = \frac{2\mathrm{NBA}}{\Delta \mathrm{t}}$, where N is number of turns in the coil.

The current in the coil will be $i = \frac{e}{R} = \frac{1}{R}\frac{2NBA}{\Delta t}$

where R is the resistance of the circuit.

The current persists only during the change of flux i.e. for the time interval $\Delta \mathrm{t}$ second.

So, the charge passed through the circuit is $\mathrm{q} = \mathrm{i} \times \Delta \mathrm{t} = \frac{2\mathrm{NBA}}{\mathrm{R}}$.

Here $\mathrm{N} = 500,\ \mathrm{B} = 0.2$ weber $/$ meter ^{2}, $\mathrm{A} = 4.0\ \mathrm{cm}^2 = 4.0 \times 10^{-4}$ meter ^{2} and $\mathrm{R} = 50\mathrm{ohm}$.

$\therefore$ $\mathrm{q} = \frac{2\times500\times0.2\times4.0\times10^{-4}}{50} = 1.6 \times 10^{-3}$ **coulomb.**

Q.6 Answer:

$\pi \times 10^{-9} \cos \omega t$ weber.

Solution:

The magnetic field at the centre of the larger loop of radius a is $B = \frac{\mu_0 i}{2a} = \frac{\left(4\pi\times10^{-7}\right)\times1}{2\times0.1} = 2\pi \times 10^{-6}$ weber $/\mathrm{m}^2$.

This field is perpendicular to the plane of the loop. The instantaneous magnetic flux linked with the smaller loop (area A) placed at the centre of the larger loop is $\Phi = \overrightarrow{\mathrm{B}} \cdot \overrightarrow{\mathrm{A}} = \mathrm{BA}\cos\omega\mathrm{t} = \left(2\pi \times 10^{-6}\right) \times \left(5 \times 10^{-4}\right)\cos\omega\mathrm{t} = \pi \times 10^{-9}\cos\omega\mathrm{t}$ weber.

Q.7 Answer:

Passes through a maximum

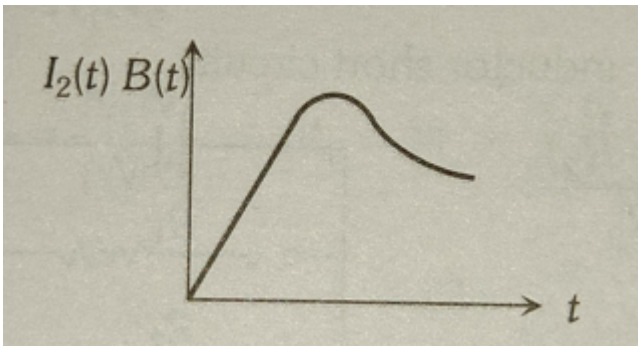
Solution:

Using k_1, k_2 etc, as different constants.

$$I_1(t) = k_1\left[1 - e^{-t/\tau}\right], B(t) = k_2 I_1(t)$$

$$I_2(t) = k_3\frac{dB(t)}{dt} = k_4 e^{-t/\tau}$$

$$\therefore I_2(t)B(t) = k_5\left[1 - e^{-t/\tau}\right]\left[e^{-t/\tau}\right]$$



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This quantity is zero for $t = 0$ and $t = \infty$ and positive for other values of t. It must, therefore, pass through a maximum.

Q.8 Answer:

$1\mathrm{mV}$

Solution:

The induced emf, $\mathrm{E} = \mathrm{B/v} = 0.2 \times 10^{-4} \times 1 \times 180 \times 1000/3600 = 0.2 \times 18/3600 = 1 \times 10^{-3}.\mathrm{V} =$
$1\mathrm{mV}$

Q.9 Answer:

$|\varepsilon|$ increases with t for $m > 1$

Solution:

$|\varepsilon| = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = \frac{-\mathrm{d}}{\mathrm{dt}}\left(\mathrm{Kt}^{\mathrm{m}}\right) = -\mathrm{Kmt}^{\mathrm{m}-1}$

From this, it follows that (1) and (2) are correct.

Q.10 Answer:

The magnetic field is in the same plane as the circular coil and it may or may not vary

Solution:

When circular coil expands radially in a region of magnetic field such that the magnetic field is in the same plane as the circular coil or the magnetic field has a perpendicular (to the plane of the coil) component whose magnitude is decreasing suitably in such a way that the cross product of magnetic field and surface area of plane of coil remain constant at every instant.

Q.11 Answer:

Just after closing the switch S, no current flows in 2Ω.

Solution:

(i) When S closed : At $t = 0$, L-open, hence no current in 2Ω.

In steady state, L offers zero resistance. Hence current I through the cell $= \frac{6}{R_{eff}}$,

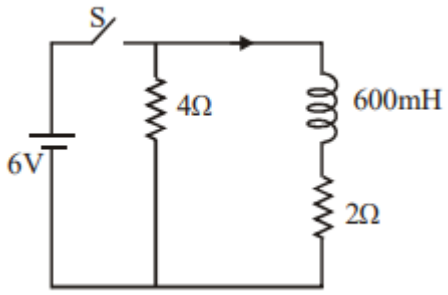
$$\mathrm{R_{eff}} = \frac{8}{6} = \frac{4}{3}\Omega \quad \therefore \quad \mathrm{I} = \frac{6\times3}{4} = \frac{18}{4} = 4.5\ \mathrm{A}$$
$$\text{Current through } 4\Omega = 4.5 \times \frac{2}{4+2} = \frac{9}{6} = \frac{3}{2} = 1.5\ \mathrm{A}$$

(ii) When S opened : Time constant

$$\tau = \frac{\mathrm{L}}{\mathrm{R}} = \frac{0.6}{4+2} = 0.1\ \mathrm{s}$$

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Just after opening S, current is steady value $I = 4.5\ A$



Q.12 Answer:

$2.4\ m/s^2$

Solution:

$i = \frac{9}{3} = 3amp.; \quad a = \frac{F}{m} = \frac{Bi\ell}{m} = \frac{0.8\times3\times1}{1} = 2.4\ m/s^2$

$e = Bv\ell$

when $e = 0.8 \times 6 \times 1 = 4.8\ V$

when $e = 0.8 \times 6 \times 1 = 4.8\ V$

$i = \frac{9-4.8}{3} = \frac{4.2}{3}; a = 0.8 \times \frac{4.2}{3} = 1.1\ m/s^2; a = 0,\ F_{net} = 0$

$9 = Bv_T\ell$

$v_T = \frac{9}{0.8\times1} = 11.25\ m/s$

Q.13 Answer:

$1.1\ m/s^2$

Solution:

$$i = \frac{9}{3} = 3amp.; \quad a = \frac{F}{m} = \frac{Bi\ell}{m} = \frac{0.8 \times 3 \times 1}{1} = 2.4\ m/s^2$$

$$e = Bv\ell$$

when $e = 0.8 \times 6 \times 1 = 4.8V$

$i = \frac{9-4.8}{3} = \frac{4.2}{3}; a = 0.8 \times \frac{4.2}{3} = 1.1\ m/s^2; a = 0,\ F_{net} = 0$

$$9 = Bv_T\ell$$

$$v_T = \frac{9}{0.8 \times 1} = 11.25\ m/s$$

Q.14 Answer:

$11.25\ m/s$

Solution:

$i = \frac{9}{3} = 3amp.; \quad a = \frac{F}{m} = \frac{Bi\ell}{m} = \frac{0.8\times3\times1}{1} = 2.4\ m/s^2$

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$9 = Bv_T\ell$

$v_T = \frac{9}{0.8\times1} = 11.25\ m/s$

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Q.15 Answer:

$0.675\ \mathrm{V}$

Solution:

(i) $\varepsilon = \mathrm{vBL} = (2.00\ \mathrm{m/s})(0.450\ \mathrm{T})(0.300\ \mathrm{m}) = 0.675\ \mathrm{V}$
(ii) $\mathrm{E} = \frac{\mathrm{V}}{\mathrm{L}} = \frac{0.675\ \mathrm{V}}{0.300\ \mathrm{m}} = 2.25 \frac{\mathrm{V}}{\mathrm{m}}$
(iii) The positive charges are moved to end b, so b is at a higher potential.

Q.16 Answer:

$2.25\ \mathrm{V/m}$

Solution:

(i) $\varepsilon = \mathrm{vBL} = (2.00\ \mathrm{m/s})(0.450\ \mathrm{T})(0.300\ \mathrm{m}) = 0.675\ \mathrm{V}$
(ii) $\mathrm{E} = \frac{\mathrm{V}}{\mathrm{L}} = \frac{0.675\ \mathrm{V}}{0.300\ \mathrm{m}} = 2.25 \frac{\mathrm{V}}{\mathrm{m}}$
(iii) The positive charges are moved to end b, so b is at a higher potential.

Q.17 Answer:

Both (A) and (B)

Solution:

(i) $\varepsilon = \mathrm{vBL} = (2.00\ \mathrm{m/s})(0.450\ \mathrm{T})(0.300\ \mathrm{m}) = 0.675\ \mathrm{V}$
(ii) $\mathrm{E} = \frac{\mathrm{V}}{\mathrm{L}} = \frac{0.675\ \mathrm{V}}{0.300\ \mathrm{m}} = 2.25 \frac{\mathrm{V}}{\mathrm{m}}$
(iii) The positive charges are moved to end b, so b is at a higher potential.

Q.18 Answer:

$(A) - p, s, (B) - q, r, s (C) - p, s (D) - p, s$

Solution:

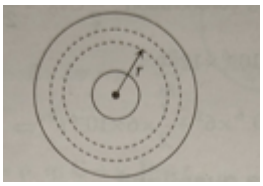
(A) $-$ **ps,** $(B)-$ **qrs,** $(C)-$ **ps, (D)** $-$ **ps.**
(A) Instantaneous $\mathrm{E} = \mathrm{B}\ell\mathrm{v} = \mathrm{Byv} \Rightarrow \mathrm{E} = \mathrm{f(y)} \Rightarrow$ **sinusoidal of** t **maximum value of** $E = By_{\max} V = Bav$
(B) No change in area $\Rightarrow \mathrm{E} = 0$
(C) $\phi = \overrightarrow{\mathrm{B}} \cdot \overrightarrow{\mathrm{A}} = \mathrm{BA}\cos\theta = \mathrm{BA}\cos\omega\mathrm{t}$, **neglecting initial phase and sign,** $\mathrm{E} = \frac{\mathrm{d}\phi}{\mathrm{dt}} = \mathrm{BA}\omega\sin\omega\mathrm{t}$
Here, $\omega = \mathrm{b} \quad \therefore$ **sinusoidal with maximum value** $= \mathrm{BAb}$ **Area** $\mathrm{A} = \int_0^{\pi/\mathrm{b}} \mathrm{a}\sin\mathrm{bx}\mathrm{dx} = -\frac{\mathrm{a}}{\mathrm{b}}\cos\mathrm{bx}\Big|_{\mathrm{x}=0}^{\mathrm{x}=\pi/\mathrm{b}} = \frac{-2\mathrm{a}}{\mathrm{b}}$
$\therefore \quad \mathrm{E}_{\max} = \mathrm{BAb} = 2\mathrm{Ba}$
(D) $\phi = \mathrm{B} \cdot \mathrm{A} = \mathrm{B}_0 \sin\mathrm{bt} \times \left(\frac{2\mathrm{a}}{\mathrm{b}}\right) \quad \therefore \mathrm{E} = \frac{\mathrm{d}\phi}{\mathrm{dt}} = \mathrm{B}_0\cos\mathrm{bt} \cdot 2\mathrm{a}$
sinusoidal, maximum $\mathrm{E} = 2\mathrm{aB}_0$

Q.19 Answer:

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Solution:

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The magnetic field inside is only due to the current of the inner cylinder.

$$B = \frac{\mu_0 i}{2\pi r}$$

Magnetic field energy density is not uniform in the space between the cylinders. At a distance r from the centre,

$$u_B = \frac{B^2}{2\mu_0} = \frac{\mu_0 i^2}{8\pi^2 r^2}$$

Energy in volume of element (length ℓ)

$$dU_B = u_B dV = \frac{\mu_0 i^2}{8\pi^2 r^2}(2\pi r \ell)dr = \frac{\mu_0 i^2 \ell}{4\pi}\frac{dr}{r}$$

$$U_B = \frac{\mu_0 i^2 \ell}{4\pi}\int_0^b \frac{dr}{r} = \frac{\mu_0 i^2 \ell}{4\pi}\ln\frac{b}{a}$$

Using values, we get $U = 7n$.

Q.20 Answer:

—

Solution:

$\int \vec{E} d\vec{\ell} = -A\frac{dB}{dt}$
As $B = 17 + (0.2)\sin(\omega t + \phi)$;

$$E(2\pi r) = -\pi r^2 (0.2)\omega \cos(\omega t + \phi)$$

$$E = -\frac{r}{2}(0.2)\omega \cos(\omega t + \phi)$$

Magnitude of the amplitude

$$= \frac{r}{2}(0.2)\omega = 2 \times 10^2 \text{mN/C}.$$

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